Distilling Frozen Time-Series Foundation Model Teachers into Lightweight Edge Anomaly Detectors: A Preregistered Quality–Fidelity–Cost Study

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Abstract

When a frozen time-series foundation model is distilled into a lightweight edge student, the bottleneck sits at the distillation interface, not at distillation itself. This preregistered study covers 29 preregistered flows plus one extension source, eleven arms, and 1,856 runs. Students trained on the teacher's scores transfer almost none of its ranking behavior (median Spearman $\boldsymbol{\rho = 0.184}$). All confirmatory comparisons are non-significant at the nine held-out sources. Switching to the teacher's forecasts as the target recovers most of the gap on development flows. Three teacher families agree that ranking fidelity spans near zero to 0.99 while no family gains detectable quality. Students run 200–1500× faster per point than the teacher on trained artifacts, and quantized PatchMLP students fit sub-millisecond, sub-megabyte budgets. On a Raspberry Pi 4B students measure 0.44 to 15.47 ms p95 while the teacher needs 2.4 s per point. Zero-tuning classical scorers top the held-out macro.

Keywords: time series anomaly detection knowledge distillation time-series foundation models edge deployment ONNX quantization preregistered evaluation

1 Introduction

Time-series anomaly detection (TSAD) at the edge faces a persistent trade-off. Detection quality demands expressive models. Latency, memory, and storage budgets cap what can be deployed. Time-series foundation models (TSFMs) such as Chronos, TimesFM, and Moirai add a new option. Used zero-shot through a forecasting-residual pipeline, a frozen TSFM scores anomalies on unseen streams with no per-device training. Here "edge" denotes sub-millisecond, sub-megabyte inference on exported artifacts. On-device validation on a Raspberry Pi 4B replicates the desktop protocol at the reference dimension (Sections 6.3 and 7.4).

Figure 1 sketches the deployment question. Three candidate interfaces move the frozen teacher's signal into a compact student during offline training, and only the student artifact reaches the edge device. Zero-shot audits paint a conditional picture. Frozen TSFMs are competitive on some sources, weak on others, and orders of magnitude more expensive to run than classical detectors.

This paper examines the natural next step. If the teacher is strong but heavy, distill it into a lightweight student and deploy the student at the edge. The proposal sounds straightforward, and its components exist. Knowledge distillation for anomaly detection is established, though the teacher there is domain-adapted and remains in the inference loop. TSFM compression is established on the forecasting side. What is missing is a disciplined answer to the deployment question itself. When the teacher's detection behavior is moved into a compact student ($\approx$0.27M parameters at the reference dimension, scaling with channel count) that must run without the teacher, does anything of value transfer? Is the result competitive with strong classical baselines under the same resource budget?

We answer this question with a preregistered study. We lock a causal scoring structure on a frozen Chronos-Bolt teacher, with model-specific point-forecast extraction when other teacher families are compared. We define three evaluation axes: quality measured by the volume under the precision–recall surface (VUS-PR), fidelity to the teacher's scoring behavior, and edge cost on exported artifacts. We train eleven arms on 20 development flows from source-blocked collections. These include score-level and ranking-level distillation of the teacher's scores, a forecast-distillation control, self-supervised students that never see the teacher, a linear floor, a fixed-configuration PCA baseline, and the teacher itself. We then confirm the preregistered arms at 10 seeds on 9 held-out sources. In total: 1,270 preregistered training runs across 29 flows from 22 data collections, with the full preregistration and its revision log in the artifact package.

The answer, under this protocol, is a *delineated boundary* rather than a blanket verdict, and its location is instructive. The distillation *interface* decides what transfers. Score-level distillation transfers little of the teacher's ranking behavior (median $\rho = 0.184$) and shows no statistically detectable quality advantage; the study is powered to rule out only large differences. On held-out sources the score arm is the weakest of the six student arms, behind self-supervised students and behind a linear predictor. Replacing the score target by the teacher's *forecast* closes most of the deficit on the development set and is equivalent to self-supervision on held-out sources. Four controls (shuffled-teacher, equal-context, anomaly-exposure, and a second teacher family) trace

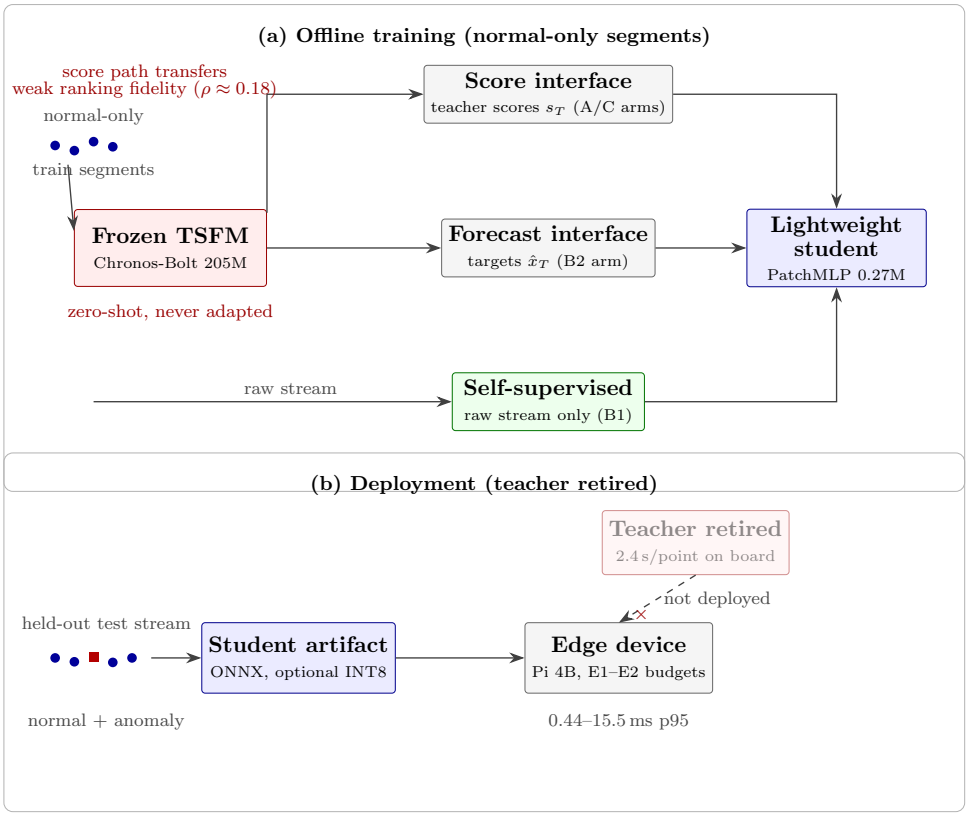


Fig. 1 Mechanism overview. (a) Offline training: a frozen TSFM teacher feeds three candidate interfaces (its anomaly scores, its point forecasts, or no teacher signal at all) into a compact student trained on normal-only segments. (b) Deployment: only the exported student artifact runs on the edge device; the teacher is retired and never deployed. The score interface transfers little ranking fidelity.

the failure to the score interface itself. The teacher is simultaneously 200–1500× slower per point than its students and no better than the strongest of them on held-out sources. The fixed-configuration PCA baseline tops the macro table on certain anomaly morphologies but fails catastrophically on others that the students handle. Three zero-tuning classical scorers added under the final revision sharpen the classical anchor: two of them top the held-out macro above every neural arm, while trained students keep a clear advantage on the highest-channel source. What remains valuable is conditional: score-distilled students occupy a niche on persistence-type anomalies where both the teacher's signal and temporal structure matter.

Our contributions are fourfold. (C1) A preregistered, source-blocked evaluation protocol for TSFM-teacher distillation into edge students, covering quality, fidelity, and deployment cost on exported ONNX artifacts with optional INT8 quantization and explicit E1/E2 edge budget tiers. (C2) A mechanism-level boundary delineation. The

distillation interface, not distillation itself, decides what transfers. Score-level distillation transfers little fidelity and no detectable quality advantage over self-supervision. (C3) The interface analysis itself. The teacher's *forecast* target is equivalent to self-supervision while its *score* target is not, with shuffled-teacher, equal-context, anomaly-exposure, and second-teacher controls attributing the failure to the score interface. (C4) A practitioner selection table mapping anomaly morphologies and resource budgets to detectors. All artifacts will be released with the submission and archived upon acceptance.

2 Related Work

2.1 Knowledge Distillation for Anomaly Detection

Knowledge distillation (KD) [6] entered anomaly detection as a *detection mechanism*: a pre-trained teacher and a student of identical capacity are both trained on normal data, and test-time anomalies are flagged by the divergence between their outputs. This teacher–student paradigm is established in the image domain [7] and, for time series, AnomalyLLM [12] adapts a large-language-model teacher to the target domain and scores anomalies by teacher–student divergence, claiming the first KD-based time-series anomaly detector. Multi-teacher KD for anomaly detection [14] extends the paradigm in the Journal of Systems Architecture, and attention-based TSAD frameworks such as IA-AD appear in The Journal of Supercomputing [10]. These detectors are single-model systems trained per domain, not distilled edge students with a teacher-exit deployment path. Our setting differs on all three defining axes. First, the teacher is a *frozen* time-series foundation model (TSFM) used *zero-shot*: it is never adapted to the target domain, and its anomaly scoring pipeline is fixed in advance. Second, distillation serves as *model compression* rather than as the scoring mechanism. The student replaces the teacher entirely at inference time, whereas mechanism-style KD keeps the teacher in the inference loop. Third, we evaluate the resulting trade-off along quality, fidelity, and cost axes under explicit resource budgets, which mechanism-style studies do not consider.

2.2 Compressing and Distilling Time-Series Foundation Models

On the forecasting side, compressing TSFMs is well explored: the Chronos family introduces Chronos-Bolt, an efficient encoder-only variant [2]. DistilTS [9] distills TSFM forecasting capability (output and hidden alignment under difficulty-aware sample weighting) into compact students. TTM [4], SparseTSF [11], and FITS [24] build tiny forecasters from scratch. TinyCast [21] reaches 146K-parameter zero-shot probabilistic forecasting with computed periodicity and depthwise-separable convolutions, without distillation. Concurrent work explores the opposite direction, keeping the TSFM in the inference loop: ChronosAD [8] uses Chronos as a zero-shot feature extractor with a temporal refinement head, at the cost of a foundation model at inference time. On the classification side, TinyCN [15] distills the Mantis foundation model [5] into tiny convolutional classifiers. None of these works distills *detection* behavior, and a forecaster

distilled for point accuracy need not preserve anomaly-scoring behavior; we therefore treat "forecast-distillation-then-residual" as an explicit control arm (B2) rather than assuming it equivalent to score distillation. Concurrent domain-specific work distills fine-tuned industrial foundation models for particle-physics trigger hardware [1]; it neither addresses general-purpose anomaly detection nor uses frozen zero-shot teachers.

2.3 TSFMs as Zero-Shot Anomaly Detectors

Systematic evaluations show that frozen TSFMs can act as zero-shot anomaly detectors through forecasting-residual pipelines, but with strongly condition-dependent reliability. Aggregated evaluations over public benchmarks [20] and the TSB-AD suite [13] report large variance across datasets, and continuous (level-shift) anomalies are a known failure signature [23]. A leakage-aware evaluation protocol with a lightweight trainable head over frozen TSFM representations [25] reaches conclusions consistent with representation–detectability gaps. These studies characterize *how good the teacher is*; none asks whether that behavior can be moved into a deployable student. Our protocol reuses their interface design (Section 3) and cites their numbers as teacher priors, while the fidelity–quality analysis is complementary to their audit perspective.

2.4 Lightweight Time-Series Anomaly Detection

Lightweight TSAD spans classical baselines (Matrix Profile [26], PCA reconstruction, LOF, one-class SVM), compact neural detectors, and edge-oriented systems that combine KD with quantization [7]. Benchmark suites score such methods on accuracy, with TSB-AD [13] and TAB [19] using VUS-PR [17] as the primary metric, but do not couple accuracy to latency, model size, or memory budgets, and edge-oriented TSAD papers typically report custom efficiency metrics. The combination of a fixed accuracy benchmark with explicit resource-constraint tiers and a preregister protocol is, to our knowledge, open territory; our three-axis evaluation framework is designed to occupy it.

2.5 Positioning

In summary, existing work provides (i) the KD-AD lineage whose teachers are domain-adapted and remain in the inference loop, (ii) TSFM compression for forecasting and classification, (iii) audits of frozen TSFMs as detectors, and (iv) accuracy-only lightweight benchmarks. What is missing is a preregistered answer to the deployment question. When a frozen TSFM's detection behavior is distilled into a compact student that must run without the teacher, what actually transfers, and is the result competitive with strong classical baselines under the same resource budget? Section 3 defines our protocol for answering this, and Sections 5 and 6 report the answer, including a prominent boundary delineation. The score interface fails to transfer, the forecast interface does not.

3 Preregistered Protocol

The protocol was preregistered (v1.0, with an append-only revision log R1–R5 disclosed in Section 3.4 and the full hash chain in Appendix A) *before* any test-segment metric was computed; the confirmation-set arm list was amended in R5 based on dev-set evidence only, with the ruling preceding the launch of the confirmation runs and the confirmation data untouched throughout. This section states the locked decisions; datasets, implementation details, and the statistical protocol are specified in Section 4.

3.1 Detection Interface (Teacher Lock)

The teacher is a frozen TSFM combined with a fixed, causal detection pipeline: (i) context-wise z-normalization using statistics of the context window only; (ii) one-step probabilistic forecasting of x_t from $x_{<t}$, with the *median* quantile taken as the point forecast (uniform across model families); (iii) per-channel absolute residuals $|x_{t,c} - \hat{x}_c(t)|$; (iv) cross-channel aggregation by *max*. The teacher score is therefore $s_T(t) = g(x_{\leq t})$; quantile-interval scoring is deliberately excluded (interface sensitivity belongs to our companion audit). Context length is each model's library default and is recorded in the cache metadata. Distillation aligns the student to exactly this interface. The student input window ends at the same timestamp the teacher scores, so student and teacher scores are timestamp-aligned by construction. After training, the teacher is removed from the deployment path entirely, and the student is the only inference-time artifact. Input windows are leakage-audited by arm: score-distilled students (A/C) receive the window $x_{t-L+1..t}$ *including* the current observation x_t, whereas the regression arms (B1/B2) receive $x_{t-L..t-1}$ and predict x_t, so the prediction target never enters the input. The teacher scores with a 512-point context and the students with 128-point windows; the interface ablation at L=512 (R7, exploratory, five flows) does not support this asymmetry as the driver of the ladder conclusion (four of five flows), though full information-set equalization across all arms remains future work. Figure 2 summarizes the locked design.

3.2 Three-Axis Evaluation Framework

All arms are evaluated on three axes. **Quality** is VUS-PR (primary) and VUS-ROC (secondary), never using point adjustment or thresholded F_1. **Fidelity** measures how faithfully the student reproduces the teacher's scoring behavior (full-series and tail-restricted rank agreement); it is a diagnostic axis, reported even when it conflicts with quality. **Cost** is measured on exported inference artifacts (ONNX, optional INT8), never on training checkpoints, under a single-thread deployment protocol with full environment disclosure. Formal definitions, tools, and protocols are tabulated in Table 1.

3.3 Arms and Controls

The preregistered arms are: the **score arm (A)**, regressing a Huber loss to the teacher score s_T; the **ranking arm (C)**, a RankNet-style objective [3] over sampled within-stream pairs (relational distillation in the spirit of RKD[18]); the **self-supervised**

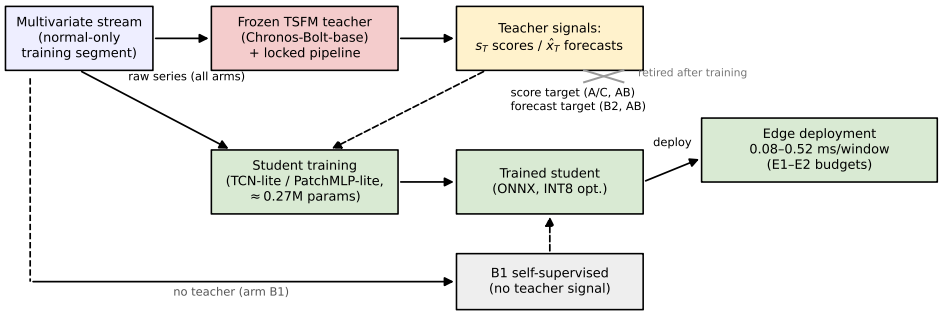


Fig. 2 Study design. The frozen TSFM teacher (Chronos-Bolt-base behind the locked causal scoring interface of Section 3.1) provides two candidate distillation signals to the student training stage: its anomaly scores s_T (arms A/C and the AB hybrid) and its point forecasts $\hat{x}_T$ (arm B2 and AB). Arm B1 receives no teacher signal and trains on the raw stream only. After training, the teacher is retired. Only the exported student artifact (ONNX, optional INT8) is deployed and timed.

control (B1), training the same backbone by next-step regression *without any teacher signal*; the **forecast-distillation control (B2)**, regressing the teacher's point forecast (DistilTS-style); the **hybrid (AB)**, coupling the score objective with a λ-weighted forecast-alignment term; a **linear floor**; and fixed references, namely the frozen teacher (T0) and a fixed-configuration PCA-reconstruction baseline fitted on the training segment only. Causal Matrix Profile and LOF were designated as auxiliary classical baselines. Causal Matrix Profile was ultimately not executed (deviation disclosed in R5): a faithful causal implementation with the full tuning budget was judged disproportionate to its marginal value. LOF was later added in a window-feature form under revision R11, fit on the full unlabeled test segment (transductive, non-streaming), alongside two zero-tuning streaming scorers. Full arm definitions, the comparison table, and implementation details are in Section 4.2.

3.4 Preregistration and Staging

The study proceeds in two source-blocked stages: an *exploration* stage on 20 development flows (arm selection, ablations, mechanism controls) and a *confirmation* stage on 9 held-out flows from source-blocked collections (one disclosed NAB exception), at increased seed count (3 seeds per arm on development auxiliary arms, 10 on the confirmation stage). The preregistration was amended five times through the append-only log (R1–R5), each before the affected measurements and each disclosed with its trigger in the revision history; three further revisions (R6–R8) registered exploratory controls after the confirmation stage began, and are flagged as such in the hash chain (Appendix A). The confirmatory analyses are exactly the three preregistered pairwise comparisons of the confirmation stage, and all other analyses (macro rankings, conditional morphology analysis, the selection table, and the shuffled-teacher control) are exploratory and labelled as such. Sibling-project reuse of teacher caches is disclosed,

Table 1 Evaluation metrics and protocols per axis. VUS-ROC and the tail-agreement companions are computed and shipped with the per-run artifacts; the paper text reports VUS-PR, ρ, and top-5% overlap.

Axis	Metric	Tool / protocol	Notes
Quality	VUS-PR (primary), VUS-ROC (secondary; artifact-only)	official VUS implementation, default buffering	no point adjustment; no thresholded F_1
Fidelity	Spearman ρ; top-1%/5% overlap; pairwise agreement; tail rank agreement (artifact-only)	per-window scores vs. teacher scores	reported even when conflicting with quality
Cost	p50/p95 latency per window; file size; peak RSS	single-thread ORT, batch 1; matrix re-timed on trained artifacts, 5 rounds $\times$ (50 warm-up + 500 timed), Appendix C	cold-start excluded (disclosed)
Equivalence	TOST-style interval reading	preregistered margin 0.05 VUS-PR	bootstrap CIs, 5,000 resamples

as is the fact that the teacher pipeline configuration is shared with a companion audit study.

4 Experimental Setup

4.1 Data: Flows and Source Blocking

The evaluation grid (Table 2) covers 29 preregistered flows drawn from 22 original data collections, split into 20 development flows (exploration and arm selection) and 9 confirmation flows whose sources are source-blocked from the development set. Labels are the official protocol labels of each collection throughout; flows whose test segments contain no anomalies are excluded from VUS computation and reported as such. Multivariate flows are reduced to per-window scores by the teacher aggregation (Section 3.1); univariate flows are scored directly. Every flow trains on its unlabeled training segment and is evaluated on its disjoint test segment; source-block splits keep the two stages disjoint with one disclosed exception (the NAB collection contributes streams to both stages). One further caveat is disclosed: SWaT and WADI share an operator testbed lineage, and the NAB collection contributes several streams to the development set while one NAB-derived stream appears in confirmation.

4.2 Arms and Reference Baselines

Table 3 summarizes the compared arms, their training signals, and their inference-time footprint. All neural arms share the two compact ($\approx$0.27M-parameter) backbones;

Table 2 Evaluation flows. Dev=development (exploration and arm selection); Conf=confirmation (held-out sources). Channels are the per-window input dimensionality; test points and AR (point-label anomaly ratio) refer to the evaluated test segment.

Stage	Flow	Collection	Channels	Test points	AR
Dev	SWaT (SWaT-id1)	SWaT testbed	66	5,474	0.363
Dev	HAI 2.2	HAI	86	86,400	0.010
Dev	SMD machine-4	Server Machine Dataset	38	18,096	0.024
Dev	SMD machine-20	Server Machine Dataset	38	6,123	0.026
Dev	PSM	Pooled Server Metrics	25	87,752	0.278
Dev	MSL-10 / MSL-5	NASA MSL	55	1,510 / 3,804	0.087 / 0.111
Dev	SMAP-4 / SMAP-13	NASA SMAP	25	2,191 / 1,943	0.055 / 0.026
Dev	GHL-2	Gas Heating Loop	19	149,000	0.012
Dev	SVDB-2	MIT-BIH SVDB (ECG)	2	175,418	0.049
Dev	EMG	EMG (Lobov et al.)	8	130,900	0.057
Dev	J-D service-0	JumpStarter KPI	38	23,705	0.004
Dev	NAB cpu-24ae8d / cpu-53ea38	NAB AWS	1	2,016 each	0.115 / 0.319
Dev	NAB occupancy-6005	NAB realTraffic	1	1,190	0.001
Dev	NAB machine-temp	NAB realKnownCause	1	11,348	0.280
Dev	Exathlon-1	Exathlon	18	30,476	0.178
Dev	Opportunity	OPP wearable	248	1,375	0.240
Dev	GECCO	GECCO water quality	9	55,409	0.004
Conf	WADI (A1 train / A2 attack)	WADI	116	172,803	0.058
Conf	Genesis	JSA Vol. 143 slice	18	682	0.073
Conf	Daphnet	Daphnet wearable	9	18,042	0.128
Conf	MGAB	MGAB synthetic	1	50,000	0.003
Conf	Yahoo-A3	Yahoo A3 benchmark	1	840	0.007
Conf	IOPS	IOPS KPI	1	74,565	0.005
Conf	MITDB-5	MIT-BIH Arrhythmia (ECG)	2	63,087	0.007
Conf	NEweather	NEweather	8	9,079	0.326
Conf	NAB exchange-2-cpm	NAB realAdExchange	1	812	0.201

the linear floor and PCA serve as non-neural references. The frozen teacher (T0) is a reference only and never participates in deployment.

4.3 Implementation and Training Details

Table 4 lists the training configuration; per-flow manifests with the exact values are part of the artifact package. Students are trained on a single RTX 5060 Ti (16 GB); edge timing uses single-thread ONNX Runtime 1.28 on the desktop CPU (12th-Gen Intel Core i7-12700F), with PyTorch 2.13 for training; the teacher scores on the same GPU.

Table 3 Compared arms. "Teacher at inference" indicates whether the frozen TSFM must be present when the arm produces anomaly scores on the test segment; only the deployed students and the classical references run without it.

Arm	Training signal	Backbone	Teacher at train	Teacher at inference
T0 (teacher)	— (zero-shot)	Chronos-Bolt-base	—	**yes**
A (score)	teacher scores s_T	TCN / PatchMLP	yes	no
C (ranking)	teacher score ranking	TCN / PatchMLP	yes	no
B1 (self-sup.)	next-step residuals	TCN / PatchMLP	no	no
B2 (forecast dist.)	teacher point forecasts	TCN / PatchMLP	yes	no
AB (hybrid)	scores + λ forecast align.	TCN / PatchMLP	yes	no
LINEAR (floor)	next-step residuals	linear	no	no
PCA (fixed config.)	train-segment fit (95% var.)	—	no	no

Table 4 Training and export configuration (identical across arms unless stated).

Item	Value
Window length L	128 (standard); 512 (interface ablation, R7)
Stride	2 (training window sampling)
Optimizer	Adam, lr 10^{-3}
Batch size	256 (A/C scoring arms); 256 (B1/B2); 128 (L=512 ablation)
Epochs	30, per flow, per seed
Seeds	5 (main grid); 3 (auxiliary arms); 10 (confirmation)
Backbone parameters (66-ch)	TCN-lite 272,401; PatchMLP-lite 265,617; linear 8,383
Teacher	Chronos-Bolt-base, context 512, median-quantile point forecast; TimesFM 2.5-200M, context 1024, mean point output (secondary subset)
Export	ONNX opset 17, dynamic batch, optional dynamic INT8
Edge timing	single-thread ORT, batch 1, p50/p95 over 200 windows

4.4 Evaluation Metrics and Statistical Protocol

Table 1 defines the three axes. Statistics operate at the source level. Per-flow seed-averaged scores are compared with two-sided exact Wilcoxon signed-rank tests under Holm correction across the three preregistered comparisons (raw and adjusted p both reported in Section 6.1, together with the exact test's minimum attainable p at n=9), effect sizes via matched-pairs rank-biserial correlation (on ranks of absolute paired differences, with sign counts alongside) and raw differences with source-level bootstrap 95% CIs, and vote counts. Equivalence claims use the preregistered 0.05 VUS-PR

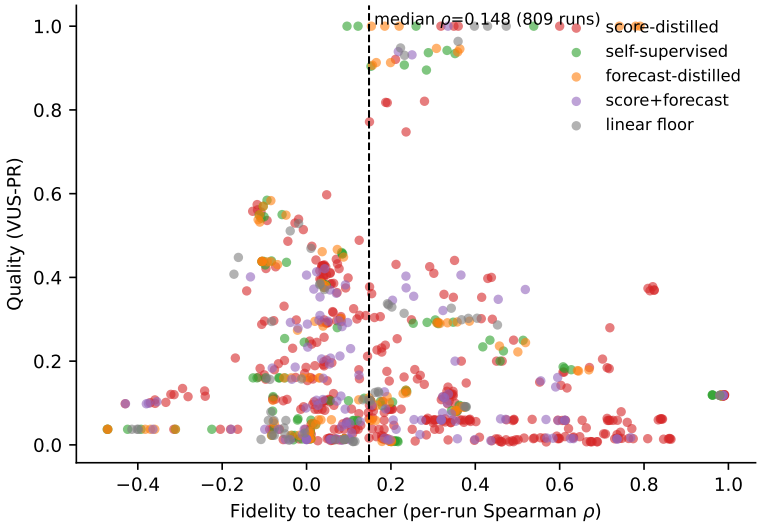


Fig. 3 Fidelity–quality decoupling. Each point is one training run with finite fidelity (809 runs: 389 main-grid plus 420 auxiliary-arm runs; per-run Spearman ρ against the teacher vs. that run's VUS-PR), colored by distillation-signal family. The dashed line is the median over all plotted runs ($\rho = 0.148$); the median over the 389 main-grid runs alone is 0.184 (Section 5.2). High quality does not require high fidelity; where fidelity is high, students inherit the teacher's level, including its weaknesses.

margin (TOST-style interval reading on the 95% CI, which is more conservative than the standard 90% TOST band). Exploratory analyses that resample at the flow level (e.g., the shuffled-teacher control) are annotated as such and do not feed the source-level claims. Any claim of "retaining teacher performance" requires this margin with interval support. Negative results are reported in full. For clarity, the *confirmatory* analyses are exactly the three preregistered pairwise comparisons of the confirmation stage; macro rankings, the conditional morphology-level analysis, and the practitioner selection table are *exploratory* and labelled as such wherever they appear.

5 Results on Development Flows

RQ1 (quality). Does the teacher improve detection quality over students trained without it, and does the answer depend on which teacher signal is distilled? **RQ2 (fidelity).** Does score distillation transfer the teacher's scoring behavior? **RQ3 (attribution).** Is any failure attributable to the score interface itself, rather than to context asymmetry, teacher capacity, or the training constraint?

This section reports the exploration stage on the 20 development flows: the preregistered main grid (arms A/C × two backbones × 5 seeds), the auxiliary arms at 3 seeds (B1, B2, AB, LINEAR), and the fixed references (frozen teacher T0, PCA). All 820 training runs completed without failure; per-flow seed dispersion is included in the artifact package.

5.1 The Full Picture

Table 5 reports VUS-PR for eleven arms, the teacher, and PCA across all development flows. Three observations structure everything that follows. First, *no method dominates*: the best per-flow arm changes across flows, and the fixed-configuration PCA baseline (the preregistered strong classical reference) attains the highest macro mean (0.286) and median (0.149) while failing catastrophically on specific flow types. The gas-heating loop is the sharpest single demonstration of the interface effect in our data. On this one flow, the teacher, every score-distilled student (0.008–0.013, including the AB hybrids), and PCA (0.007) all collapse, while the two forecast-trained students reach 0.38–0.44. Second, *eight of the ten neural student arms exceed the teacher's macro mean* (the exceptions are the two TCN variants trained on the teacher's weakest signal, A-TCN 0.137 and AB-TCN 0.130 vs.0.165); the best per-flow student beats the teacher on 16 of 20 flows (2 losses, 2 ties). The teacher is a weak spot on its own grid, echoing the condition-dependence reported by zero-shot audits. Third, the *interface* of the distillation signal matters more than the presence of a teacher. Self-supervised and forecast-distilled students (B1/B2/LINEAR, 0.217–0.253 macro) dominate score- and ranking-distilled students (A/C, 0.137–0.180).

5.2 Fidelity–Quality Decoupling

If distillation transferred the teacher's scoring behavior faithfully, the student's scores would correlate strongly with the teacher's. The data refute this. Across the 389 main-grid runs with finite fidelity (of 400; the remainder yield constant score vectors), the median teacher–student Spearman ρ is 0.184; only 22% of runs exceed $\rho > 0.5$, while 53% fall below 0.2 (Figure 3). The decoupling is flow-structured rather than noise. On the three flows where the median ρ is highest (SVDB $\rho\approx0.84$, MSL-5 $\rho\approx0.70$, J-D $\rho\approx0.61$), students track the teacher closely and inherit its level of performance. On the flows where students improve most over the teacher (e.g., SWaT: student 0.54 vs. teacher 0.33 with $\rho\approx-0.10$; NAB-occupancy: 0.56 vs. 0.20; machine-temperature: 0.43 vs. 0.33), the student is *not* imitating the teacher at all. The mechanism is consistent with these observations, which we state as a hypothesis rather than a proof. Under the normal-only training constraint, the teacher's scores act as a noisy, weakly informative regression target, and the measured gains come from the student's inductive bias filtering this target, a regularization effect, rather than from behavioral transfer. The preregistered fidelity companions agree. Across the same 389 runs, top-5% score-set overlap averages 0.242 (median 0.151; chance is 0.05) and pairwise agreement on non-tied teacher pairs averages 0.594, above chance but far from imitation. Anomaly-tail fidelity tells the same story from the deployment angle and is reported per flow in the artifact package.

5.3 The Interface Ladder

The preregistered three-level comparison asks *which distillation objective is strongest.* Ranking the objectives by their information content about the anomaly (ground-truth-free self-supervision B1, the teacher's *forecast* B2, and the teacher's *score* A/C)

Table 5 Development-set VUS-PR (mean over seeds) for eleven arms, the frozen teacher (T0), and the preregistered PCA baseline across the 20 dev flows. Best per flow in bold.

Flow	C-PM	C-TCN	A-PM	A-TCN	B1-TCN	B1-PM	B2-TCN	B2-PM	AB-PM	AB-TCN	LINEAR	T0	PCA
D10_GHL_id2	0.008	0.008	0.011	0.012	0.438	0.435	**0.438**	0.435	0.013	0.013	0.383	0.009	0.007
D11_SVDB_id2	0.061	0.058	0.060	0.059	0.081	**0.092**	0.081	0.091	0.061	0.059	0.091	0.067	0.069
D12_EMG	0.119	0.116	0.117	**0.119**	0.118	0.119	0.118	0.119	0.118	0.116	0.118	**0.119**	0.071
D13_JD1_svc0	0.014	0.019	0.015	0.016	0.009	0.008	0.009	0.009	0.015	0.016	0.008	0.073	**0.695**
D14_NAB_cpu24ae8d	0.140	0.226	0.202	0.196	0.183	0.236	0.178	0.234	0.201	0.202	**0.305**	0.222	0.136
D15_NAB_cpu53ea38	0.304	0.292	0.304	0.298	0.294	0.291	0.292	0.293	0.288	0.297	0.305	0.281	**0.330**
D16_NAB_occ6005	0.560	0.108	0.550	0.018	**1.000**	0.483	**1.000**	**1.000**	0.516	0.017	**1.000**	0.200	0.036
D17_NAB_machtemp	0.297	0.397	0.350	0.427	0.457	0.260	**0.459**	0.286	0.344	0.396	0.334	0.328	0.284
D18_Exathlon_id1	0.481	0.550	0.273	0.117	0.915	0.925	0.933	0.922	0.656	0.100	0.947	0.561	**0.999**
D19_OPP_id3	0.202	0.257	0.180	0.214	0.160	0.160	0.160	0.160	0.176	0.226	0.158	0.255	**0.439**
D1_SWaT_id1	0.542	0.524	0.424	0.396	0.545	**0.568**	0.540	0.567	0.384	0.350	0.503	0.331	0.446
D20_GECCO_id1	0.015	0.014	0.011	0.014	0.014	0.014	0.014	0.014	0.013	0.014	0.013	0.014	**0.403**
D2_HAI	0.013	**0.018**	0.014	0.014	0.016	0.013	0.015	0.015	0.014	0.013	0.012	0.015	0.013
D3_SMD_id4	0.055	0.069	0.068	0.050	0.044	0.028	0.044	0.028	**0.070**	0.048	0.032	0.069	0.040
D4_SMD_id20	0.030	0.061	0.161	0.099	0.078	0.023	0.071	0.023	0.046	0.084	0.042	0.030	**0.885**
D5_PSM	0.372	0.416	0.419	0.392	0.382	0.381	0.352	0.383	0.393	0.350	0.380	0.388	**0.501**
D6_MSL_id10	0.085	0.088	0.090	0.098	0.105	0.100	0.105	0.100	0.089	0.096	**0.107**	0.101	0.080
D7_MSL_id5	**0.207**	0.109	0.155	0.106	0.129	0.107	0.126	0.112	0.151	0.110	0.126	0.129	0.162
D8_SMAP_id4	0.059	0.060	0.059	0.059	0.058	0.060	0.059	0.060	0.059	0.060	0.058	0.065	**0.080**
D9_SMAP_id13	**0.037**	**0.037**	**0.037**	**0.037**	**0.037**	**0.037**	**0.037**	**0.037**	**0.037**	**0.037**	**0.037**	**0.037**	**0.037**
Macro mean	0.180	0.171	0.175	0.137	0.253	0.217	0.251	0.244	0.182	0.130	0.248	0.165	0.286
Median	0.102	0.098	0.136	0.099	0.123	0.113	0.122	0.115	0.104	0.090	0.122	0.110	0.149

produces a consistent macro separation (B1/B2/LINEAR $\approx$ 0.22–0.25 vs. A/C $\approx$ 0.14–0.18), though within the upper family the B1-vs-B2 ordering swaps with the backbone (B2-TCN 0.251 edges out B1-PM 0.217's family range; on the confirmation set the same comparison reverses, Section 6.1). Score-level distillation, the interface this paper set out to test, is the weakest of the three; the pairwise per-flow record confirms it is not a macro artifact (B1-TCN beats C-PM on 11 of 20 flows). Notably, a linear predictor with residual scoring (0.248 macro) comes within 0.003 of the strongest distilled neural student on development (B2-TCN 0.251) and is the strongest student arm on held-out sources (0.323). The AB arm, which couples score distillation with forecast alignment, recovers part of the gap over pure score distillation (AB-PM 0.182 vs. A-PM 0.175 at macro) but remains far below forecast-only training (B2-PM 0.244), indicating that the alignment benefit comes from the forecast target itself rather than from the combination. An interface ablation (revision R7, exploratory) addresses the main confound directly. Retraining A-PM and B1-PM at the teacher's own context length (L=512 instead of 128) on five representative flows leaves the ladder intact. The B1$-$A difference is +0.148 at L=512 vs. +0.144 at L=128 (per-flow direction agrees on 4 of 5; the disagreement is the occupancy flow). On these five flows, the context-length asymmetry between teacher and student does not support an explanation of the interface gap.

A teacher-on-anomaly control (revision R8, exploratory) tests whether the normal-only training signal, rather than the interface itself, is the binding constraint. Synthetic anomalies (1% spikes, 1% 100-point level shifts, $k \in \{3, 5, 8\}\sigma$) are injected into the training segments of five representative flows; the injection parameters are chosen independently of the students' window-residual mechanism (spike and level-shift morphologies, not residual-shaped anomalies), so no arm is favored by construction; the teacher scores the augmented segments (its score discriminates injected from normal points on three of five flows, ratios 1.2–2.8$\times$), and A-PM and B1-PM are retrained on the augmented streams. The result is sharply bimodal (Figure 4). Where the teacher discriminates, score distillation with anomaly exposure improves dramatically (Exathlon: 0.27 $\rightarrow$ 0.86); where it does not (occupancy: teacher ratio < 1), augmented distillation degrades equally dramatically (0.55 $\rightarrow$ 0.08). The self-supervised control is robust to the same augmentation (mean change +0.001, per-flow range $-$0.002 to +0.008 across the five flows). The score interface, then, is not categorically unable to transfer detection behavior. It inherits the teacher's condition-dependent inconsistency. Deploying it is a bet on the teacher responding informatively to the anomalies at hand, and verifying the bet requires running the teacher on representative data, an offline cost that is amortized across deployments but still non-trivial compared to fully self-supervised training.

5.4 Conditional Analysis: When Is Each Arm Worth It?

Although PCA tops the macro table, its wins concentrate on flows with abrupt level/regime anomalies (e.g., Exathlon, SMD-20, J-D), where reconstruction error separates regimes almost perfectly; on temporally structured or persistence-type anomalies its failures are severe (gas-heating loop 0.007; SWaT 0.446 vs. student 0.542). Ranking- and score-distilled students occupy a complementary niche. Their

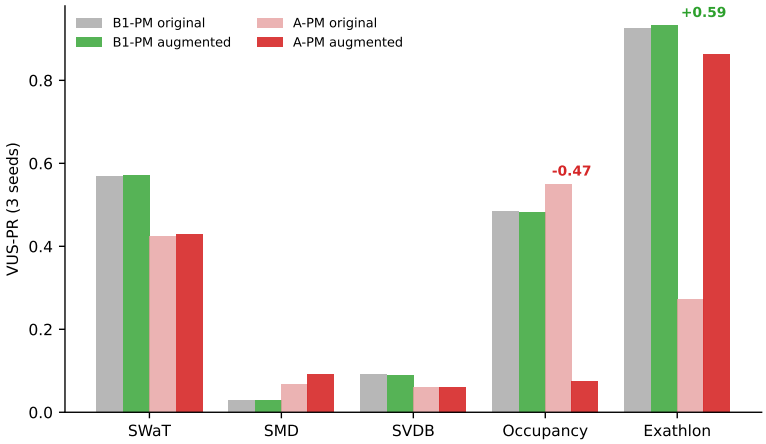


Fig. 4 Teacher-on-anomaly control (R8, exploratory): VUS-PR (mean of 3 seeds) on five development flows, four bars per flow. Gray/green: B1-PM trained on original vs. anomaly-augmented segments; light red/red: A-PM original vs. augmented. Bold annotations give the within-arm augmented-minus-original difference for A-PM (the same quantity quoted in the text). The outcome is bimodal. Where the teacher's scores discriminate the injected anomalies, augmented score distillation improves sharply (Exathlon: 0.27 → 0.86); where they do not (Occupancy, teacher ratio <1), it degrades sharply (0.55 → 0.08). B1-PM is robust to the same augmentation on all five flows.

strongest flows (SWaT 0.542; NAB-occupancy 0.561; machine-temperature 0.428 under A-TCN) are precisely those where the teacher is weak and where temporal persistence matters. Per-flow win counts summarize the picture. The best-per-flow student beats PCA on 11 of 20 flows (loses 8, ties 1) and beats the teacher on 16 (loses 2, ties 2). Section 6 tests whether this conditional structure replicates on held-out sources; Section 7 condenses it into a practitioner selection table.

5.5 Negative and Boundary Results

Reported in full per the preregistration: (i) score-level distillation does not transfer teacher behavior faithfully (median ρ 0.184) and does not beat self-supervision; (ii) the ranking objective (C) beats the score objective (A) at macro on both backbones, but the advantage is driven by two flows and flips at the median (A-PM 0.136 vs. C-PM 0.102), so we do not claim it as systematic; (iii) one development-set flow (SMAP-13) is undetectable by *every* method including the teacher (single anomaly window in the context-undefined region); (iv) the AB arm's forecast-alignment term does not close the gap to forecast-only training.

6 Confirmation Study

The confirmation stage evaluates five preregistered arms (C-PM, B1-TCN, B1-PM, B2-PM, LINEAR) plus one post-hoc addition (A-PM), each at 10 seeds on the 9 held-out flows whose sources are source-blocked from the development set, with the frozen

Table 6 Confirmation-set VUS-PR (mean over 10 seeds) on 9 held-out sources. The three preregistered pairwise comparisons are all statistically non-significant at the source level (Section 6.1); directional consistency with the development set is noted in the text. Bold marks the row maximum (references included).

Flow	A-PM	C-PM	B1-TCN	B1-PM	B2-PM	LINEAR	T0	PCA
WADI (C1)	0.048	0.051	0.041	0.062	0.064	0.061	0.052	**0.315**
Genesis (C2)	0.170	0.147	0.135	0.135	0.135	0.132	0.169	**0.959**
Daphnet (C3)	0.307	0.317	0.231	0.245	0.245	0.252	**0.416**	0.111
MGAB (C4)	0.009	0.008	**0.124**	0.011	0.010	0.016	0.009	0.009
Yahoo (C5)	0.164	0.386	0.337	**0.773**	0.749	0.754	0.301	0.242
IOPS (C6)	0.089	0.217	0.501	0.392	0.286	**0.574**	0.261	0.019
MITDB (C7)	0.008	0.008	**0.023**	0.006	0.006	0.006	0.008	0.014
NEweather (C8)	0.930	0.933	0.930	0.928	0.927	0.927	0.946	**0.952**
AdExchange (C9)	0.182	0.185	0.177	0.183	0.181	0.183	**0.227**	**0.227**
Macro mean	0.212	0.250	0.278	0.304	0.289	**0.323**	0.265	0.316

Table 7 Confirmation-set macro VUS-PR over nine held-out sources. Mean with flow-level bootstrap 95% CI (5,000 resamples) and median. The three classical baselines are zero-tuning methods added under R11; LOF-window is fit on the full unlabeled test segment (transductive, non-streaming) and is excluded from streaming-deployment claims; all other rows are preregistered arms, the teacher, or the fixed PCA baseline.

Detector	Macro mean	95% CI	Median
Frozen teacher (T0)	0.2654	[0.114, 0.459]	0.2269
Robust-Z (zero tuning)	0.2392	[0.092, 0.443]	0.1258
MA-residual (zero tuning)	0.3248	[0.146, 0.525]	0.2232
LOF-window (zero tuning)	0.3271	[0.197, 0.500]	0.2402
PCA (fixed)	0.3165	[0.106, 0.561]	0.2269
Linear floor	0.3229	[0.124, 0.545]	0.1835
B1-PM (self-sup.)	0.3039	[0.113, 0.525]	0.1835
B1-TCN (self-sup.)	0.2778	[0.125, 0.473]	0.1770
B2-PM (forecast-dist.)	0.2893	[0.102, 0.509]	0.1809
C-PM (ranking-dist.)	0.2501	[0.102, 0.447]	0.1846
A-PM (score-dist.)	0.2116	[0.075, 0.406]	0.1640

teacher (T0) and the fixed-configuration PCA baseline as fixed references (450 preregistered + 90 post-hoc = 540 student runs; all completed). The three preregistered comparisons are evaluated at the source level (n=9 paired flows).

6.1 Preregistered Comparisons

All three preregistered pairwise comparisons are statistically non-significant at the source level (n=9; exact two-sided Wilcoxon signed-rank; the exact test's minimum attainable two-sided p at n=9 is 0.0039, which bounds all reported p-values from below). We report raw p first and Holm-adjusted p across the three-comparison family in brackets. **B1-TCN vs. C-PM**: mean difference +0.028 (bootstrap 95% CI

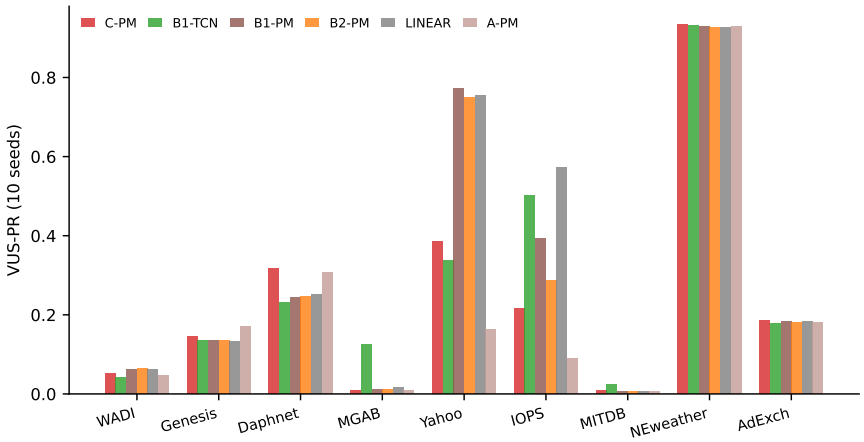


Fig. 5 Confirmation stage, per flow: VUS-PR (mean over 10 seeds) for the five preregistered student arms plus the post-hoc A-PM addition on the nine held-out sources (the same means as Table 6, which additionally appends the teacher and PCA references). No arm dominates across sources. The best arm changes from source to source; the macro ordering of the student arms is unchanged when the shared-lineage WADI source or the confirmation NAB stream is excluded.

$[-0.031, +0.103]$), p=1.0 [Holm 1.0], matched-pairs rank-biserial $r = -0.02$ (3 wins / 6 losses); the positive mean is driven by a single source (IOPS, +0.28) while C-PM wins pairwise on 6 of 9 sources. **LINEAR vs. C-PM**: +0.073 (CI $[-0.014, +0.195]$), p=0.73 [Holm 1.0], $r = +0.16$ (4 wins / 5 losses). **B2-PM vs. B1-PM**: −0.015 (CI $[-0.039, +0.000]$), p=0.20 [Holm 0.60], $r = -0.49$ (3 wins / 5 losses / 1 tie): replacing the self-supervised target by the teacher's forecast provides no detectable benefit and trends harmful, whereas on the development set the same macro comparison favored B2-PM by +0.027 (0.244 vs. 0.217), so the direction of this (non-significant) contrast is stage-dependent. A fourth, post-hoc comparison (A-PM vs. C-PM: the score vs. ranking interface; exploratory, excluded from the Holm family) is directionally consistent with the development set (−0.039, CI $[-0.097, +0.003]$, p=0.105, rank-biserial −0.62): the score arm is the weakest of the six student arms in the confirmation set and the ranking arm the weakest of the five preregistered ones, completing the score-to-ranking-to-self-supervision ordering on held-out sources. Under the preregistered 0.05 VUS-PR margin, the B2-PM vs. B1-PM 95% CI lies entirely within ±0.05, so equivalence of the two *training targets* at the mean-source level is supportable under an interval-based TOST reading (the more conservative 95% two-sided CI used here implies the 90% TOST criterion is met; Δ is the mean over the nine sources of the per-source VUS-PR difference, CIs by source-level bootstrap); the intervals for B1-TCN vs. C-PM and LINEAR vs. C-PM extend to +0.10 and +0.19 respectively, so the study is underpowered to rule out differences of that size. Table 7 consolidates the confirmation-set macro comparison with flow-level bootstrap intervals, medians, and the three zero-tuning classical baselines added under revision R11. The classical additions sharpen the headline. LOF-window (0.327) and MA-residual (0.325) land above every neural arm and the fixed PCA baseline, and the frozen teacher itself (0.265)

falls below two zero-tuning methods. The teacher-distilled arms (C-PM 0.250, A-PM 0.212) close the table's lower half, so the boundary verdict gains a classical anchor: within the tested configurations, no teacher signal we can measure survives transfer as a quality advantage over zero-tuning classical scorers. As a reference point for statistical power, with n=9 paired sources, α=0.05 two-sided, and 80% power, the minimum detectable effect is approximately $0.93\,\sigma_d$ of the paired differences, about 0.10 VUS-PR for the observed B1-TCN spread ($\sigma_d \approx 0.10$) and about 0.15 for the LINEAR spread ($\sigma_d \approx 0.16$), matching the upper range the two favorable CIs allow.

6.2 Directional Consistency and Practical Reading

Although none of the differences is significant on nine sources, the directional pattern replicates the development set. The two arms that do not use the teacher's scores, namely LINEAR and B1-PM, have the two highest macro means among the student arms (PCA, a reference, is 0.316) (macro 0.323 and 0.304 vs. 0.250 for the best score-distilled arm), the forecast-distilled arm sits between (0.289), the score arm A-PM is the weakest of the six student arms (0.212), and the exploration-stage champion C-PM is the weakest of the five preregistered arms. Two sources illustrate the conditional structure. On Yahoo the self-supervised students dominate everything (B1-PM 0.773 vs. teacher 0.301 and C-PM 0.386), whereas on WADI and Genesis the PCA baseline dominates everything (0.315 and 0.959 vs. $\leq$0.064 and $\leq$0.170 for every other method). On NEweather every method exceeds 0.92. Exclusion sensitivity. Dropping the WADI source (iTrust lineage shared with dev SWaT) or the confirmation NAB stream leaves the arm ordering unchanged (e.g., macro LINEAR/B1-PM/C-PM = 0.323/0.304/0.250 full, 0.356/0.334/0.275 without WADI, 0.340/0.319/0.258 without NAB), so neither shared-lineage caveat drives the directional pattern. A stratified reanalysis (R10) reinforces this reading. The confirmation set contains no dead flow on which every method falls below 0.02 VUS-PR, so the macro comparison is not diluted by unsignalable sources. In the four teacher-null flows the strongest teacher-free arms widen their lead (LINEAR 0.351 versus C-PM 0.165 mean). In the five sources where the teacher matches or beats the best self-supervised arm per flow (Genesis, Daphnet, MITDB, NEweather, NAB-adexchange), score distillation adds only +0.018 mean VUS-PR over that arm, while the fixed PCA baseline still dominates that subgroup (0.453). A prospective extension (revision R16) adds one fully independent tenth source (CATSv2, 153,173 test points, 5,900 anomalies). On this source every arm collapses to low absolute levels (0.036 to 0.077), score distillation shows a small edge over self-supervision (0.077 versus 0.036), and the zero-tuning LOF-window scorer dominates the source at 0.246. The primary-set ordering is unchanged on the pooled ten-source sample, and the boundary verdict gains a classical anchor on new ground. The neural arms retain one exploratory niche. On the 66-channel SWaT development source the best self-supervised student (0.568) exceeds the strongest classical baseline there (PCA at 0.446), an observation we do not extrapolate beyond this source. Figure 5 shows the per-flow results behind this analysis. The honest summary is twofold: (i) *no arm is significantly better than any other at this sample size.* The study cannot rule out differences of the size the intervals allow (up to +0.10 for B1-TCN vs.

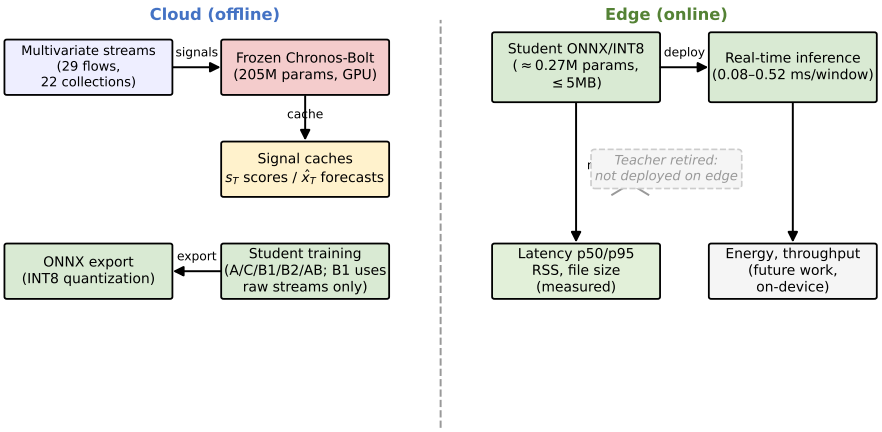


Fig. 6 Deployment split. All teacher-dependent computation, namely scoring the training segments and caching its score and forecast signals, runs offline in the cloud stage (arm B1 trains on the raw streams alone); the teacher is retired before deployment. The edge stage runs the exported student artifact alone (≈0.27M parameters, ONNX, optional INT8) and reports the measured quantities of this section (per-window latency, file size, peak RSS), on the desktop and now also on a Raspberry Pi 4B at the reference dimension, including a ten-minute sustained-throughput check; energy profiling remains future work.

C-PM and +0.19 for LINEAR vs.
C-PM); and (ii) across both stages, 29 preregistered flows from 22 original data collections, and 1,270 preregistered primary-grid runs (1,856 total including all disclosed controls and additions), *we find no evidence that a frozen TSFM teacher, through either a score or a forecast interface, improves a lightweight edge detector over self-supervised or even linear training under the same budget*, while the fidelity analysis (Section 5.2) shows the score interface does not even transfer the teacher's ranking behavior.

6.3 Cost Axis

Two budget tiers anchor the cost axis. Tier E1 requires a per-window p95 latency below 1 ms and an artifact of at most 1 MB; tier E2 extends both to 5 ms and 5 MB. Thresholds are defined on desktop single-thread measurements; the Raspberry Pi 4B figures are device-side validation, not tier definitions. The computational budget of the full study is modest. Building the Chronos-Bolt teacher cache over all thirty flows (twenty-nine preregistered plus the R16 extension) took 17.4 GPU-hours (mean 35 minutes per flow), a one-time cost shared by every distilled arm. Each student training run completes in 1–12 minutes on a single GPU (mean 84 seconds over the 450 confirmation runs, 12.5 GPU-hours in total), and the complete 1,856-run study fits within roughly 40 GPU-hours of student training plus the cached teacher passes. All cost figures below are measured on *trained* artifacts (case-C students retrained once per architecture–dimension configuration under the confirmation-stage recipe, exported to ONNX, and re-timed on an idle machine. Appendix C reports the full matrix with

round-to-round ranges). On the desktop CPU (single-thread ONNX Runtime, batch 1), the linear floor requires 0.009–0.049 ms p95 with files at or below 0.13 MB (budget tier E1 at every dimension). PatchMLP-lite runs in 0.092–0.194 ms FP32 and, after dynamic INT8 quantization, 0.058–0.116 ms with 0.22–0.47 MB files, inside E1 at every tested dimension (1 to 248 channels). TCN-lite runs in 0.414–0.653 ms FP32 (E1 at low channel counts by latency; its file crosses 1 MB from 66 channels, so E2 there), but its dynamically quantized variant is 10.3–12.2× *slower* (quantized convolution kernels in the runtime), so INT8 is recommended for the MLP-mixer architecture only. Quantization also perturbs quality. The INT8-minus-FP32 VUS-PR drift of the exported artifacts is within ±0.01 on most configurations and even reaches +0.074 and +0.097 on the 66- and 248-channel PatchMLP students (quantization noise as regularization), with one loss of −0.066 on the 66-channel linear floor (Appendix C); the linear floor, at most 0.13 MB, has no need to quantize.

The frozen teacher (Chronos-Bolt-base) is in a different cost class entirely. Figure 6 places teacher and student on opposite sides of the deployment boundary. Its offline scoring pass over the training segments of all 29 flows cost 15.4 GPU-hours (mean 32 minutes per flow), a one-time cost amortized across every student trained from the cache, but non-trivial for a single deployment and zero for self-supervised students. Measured on the same idle machine (12th-Gen Intel Core i7-12700F, ONNX Runtime 1.28, PyTorch 2.13; single CPU thread) under two matching protocols, scoring one point costs 64.7 ms at batch 64 and 122.9 ms at batch 1 (the deployment-faithful setting; 512-point context, end-to-end including normalization and aggregation). The students' end-to-end cost over full test splits (stride-1 windows, trained artifacts, two development flows and one held-out confirmation flow) is 0.07–0.51 ms p50 per window (p95 up to 0.62 ms): roughly *200–1500*× faster than the teacher at batch 1, depending on architecture and channel count. Combined with the quality results (Figure 7), the cost axis completes the picture. The teacher is simultaneously slower than every student by orders of magnitude and not better than the strongest of them on held-out sources. Peak RSS is process-level. For the sub-million-parameter students the in-process delta is below measurement noise (≤6 MB over the ORT runtime baseline), and cold-start is excluded from the per-window figures (both disclosed). The Raspberry Pi 4B on-device measurements follow the identical protocol (single-thread ONNX Runtime, batch 1, $L = 128$, CPU governor pinned to performance). On the board, the linear floor measures 0.44 ms p95, the quantized PatchMLP student 2.00 ms, and the TCN student 15.47 ms at the 66-channel reference. Process peak RSS stays at 59–62 MB with no thermal throttling (49–52°C). These board figures sit 25–33× above the matching desktop medians. The linear floor and the quantized PatchMLP student stay inside their E1 and E2 budget tiers on the board, while the FP32 TCN student lands above E2, reinforcing the recommendation to deploy the quantized MLP-mixer student. Even on the board, every student remains 8–278× cheaper than the teacher's desktop batch-1 cost of 122.9 ms. The teacher itself is not deployable on the board at all: Chronos-Bolt-base requires 2.4 s per scored point on the same device (batch 1, context 512, end-to-end), roughly 19.5× its desktop cost and 155 to 5,428× above the student artifacts.

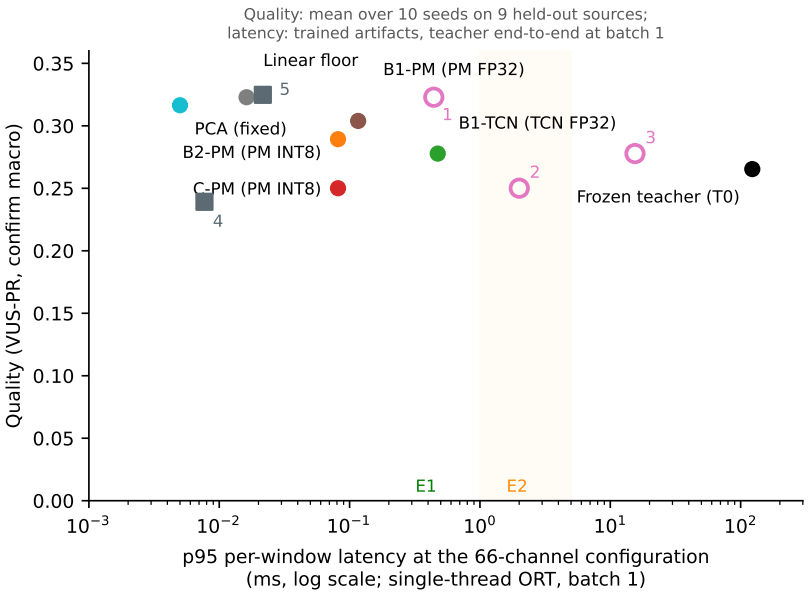


Fig. 7 Cost–quality plane on held-out sources (macro VUS-PR vs. single-thread p95 per-window latency at 66 channels, log scale; the teacher is measured end-to-end at batch 1 (deployment-faithful)). Green and orange bands are the E1/E2 latency budgets. The teacher sits in a different cost class entirely; the linear floor and the quantized PatchMLP students occupy the E1 corner. Pink open squares 1–3 mark the Raspberry Pi 4B on-device measurements of the linear floor, quantized PatchMLP, and TCN students (0.44, 2.00, and 15.47 ms p95 at the reference dimension). Gray squares 4–5 mark the zero-tuning classical scorers Robust-Z (0.008 ms) and MA-residual (0.022 ms p95, streaming, desktop single-thread).

7 Discussion

7.1 What Does TSFM Distillation Actually Transfer?

Our design cannot say what distillation transfers in general, but *here* it localizes the boundary cleanly. The interface decides. Where fidelity was high, students cloned the teacher including its weaknesses; where fidelity was near zero, students diverged from the teacher and typically beat it. Under the normal-only training constraint, the teacher's scores on clean data carry little signal that a compact forecaster cannot extract from the raw series itself. The teacher's failure modes (flat scores, persistence blind spots) are not worth copying. The forecast target tells the complementary story: B2 students, trained to predict the teacher's forecast rather than to mimic its score, close most of the gap to self-supervision on the development set and show occasional large wins. In other words, if a TSFM is in the training loop at all, our development-stage data favor its *forecasts* over its scores. Detectors trained on either target are equivalent within the preregistered 0.05 VUS-PR margin on held-out sources (the interval for the forecast-vs-self-supervision comparison lies inside the margin),

while the score interface is weaker than both. We emphasize the limitation. Equivalence of the two *training targets* is what the interval supports, not agreement of the teacher's forecasts with ground truth. A post-hoc shuffled-teacher control (revision R6, exploratory; Figure 8) sharpens the point. Retraining the score-distilled student with the teacher's scores randomly permuted in time (preserving their marginal distribution, destroying their temporal alignment; A-PM recipe, 2 seeds, all 20 development flows, bootstrap resampling at the flow level) changes mean VUS-PR by only +0.022 in favor of the true scores (10 wins, 9 losses, 1 tie; bootstrap 95% CI of the mean difference $[-0.027, +0.086]$, crossing zero; exploratory). The score interface, in this protocol, carries limited information beyond the marginal distribution of the teacher's scores, though the interval still allows a mean effect as large as +0.086. Two preregistered controls (R10) probe how general this boundary is. A capacity sweep over PatchMLP students at 0.27M, 0.6M, 1.6M, and 5.8M parameters (five development flows, three seeds; the largest size under revision R13) shows the score-interface deficit narrowing monotonically with capacity and closing only near six million parameters. Macro score distillation reaches 0.256 versus 0.346 self-supervised at 0.27M, 0.262 versus 0.342 at 0.6M, 0.291 versus 0.344 at 1.6M, and 0.332 versus 0.343 at 5.8M, where the gap −0.011 falls inside the preregistered 0.05 margin (paired mean difference −0.011, bootstrap 95% CI $[-0.054, +0.030]$, five flows). The closing size, however, prices itself out of the edge: the exported 5.8M student measures 23.2 MB FP32 and 5.8 MB quantized, 4.6 times the E2 storage ceiling in FP32 and just beyond it once quantized, far outside E1. Within edge-sized students, the interface, not capacity, is the binding constraint. Per-flow ranking fidelity to the teacher spans essentially its whole range in the same runs (median per-flow ρ from −0.13 to 0.99), and even at $\rho = 0.99$ the distilled student does not exceed self-supervised training. Ranking transfer, where it occurs, does not by itself buy quality.

A teacher-on-anomaly control (R8) sharpens this further. Even when the teacher is given the opportunity to demonstrate anomaly responses in the training signal, the outcome depends on whether the teacher actually discriminates those anomalies (Figure 4). This is exactly the condition an edge deployer cannot verify without running the teacher. A second teacher family (TimesFM 2.5) runs behind the same residual-scoring structure, with model-specific point forecast extraction (mean output rather than median quantile) and context length 1024 rather than 512. The extension was registered over all twenty-nine flows and executed on thirteen within the compute budget; ten are scorable. Three streams are shorter than the locked context (1024) and structurally unscorable for this teacher. The quality verdict replicates on the ten scorable flows. The TimesFM-taught student averages 0.239 against 0.372 for the strongest teacher-free arm, wins one flow in ten, and stays within ±0.03 of its Chronos-taught counterpart on seven of ten. The mechanism picture diverges in one instructive way, read here as an exploratory case study rather than a controlled strength experiment. Median ranking fidelity to the TimesFM teacher is 0.70 across 50 runs, roughly four times the Chronos-side 0.18. It reaches 0.99 on one development flow where the distilled student still does not exceed self-supervised training. Ranking transfer across teacher families is therefore teacher-dependent, and transfer alone buys no quality. The TimesFM teacher's own stream-level strength (T0 0.382 vs. Chronos 0.069 on SMD)

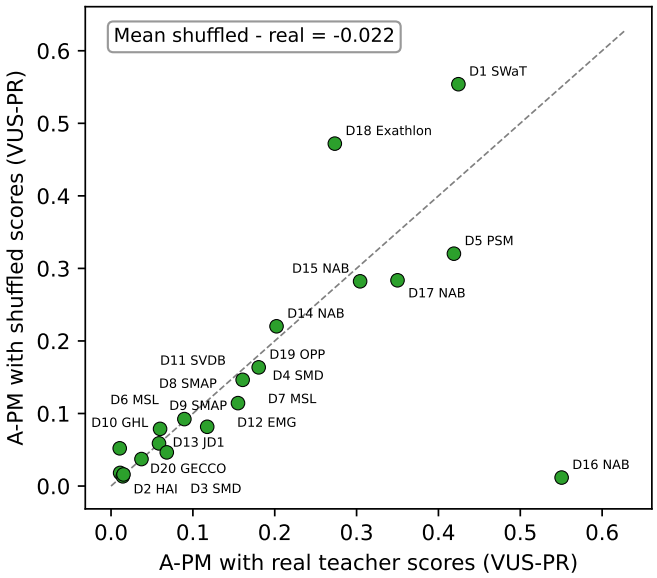


Fig. 8 Shuffled-teacher control (R6, exploratory): per-flow VUS-PR for A-PM with real teacher scores (x-axis) vs. time-shuffled teacher scores (y-axis; same marginal distribution, destroyed temporal alignment; 2 seeds per flow, paired per flow). The diagonal is $y{=}x$. Real scores win on 10 flows, lose on 9, tie on 1; mean difference +0.022, 95% CI $[-0.027, +0.086]$ (flow-level bootstrap), crossing zero.

is not converted into student advantage (0.030). A third family completes the triangle. Moirai-1.1-R-base (91M, masked-reconstruction architecture, executed under revision R16 as registered in R9) is distilled through the same score interface on the same five flows. Its macro VUS-PR is 0.317, and per-flow ranking fidelity spans -0.20 to 0.83. The instructive pattern repeats. On the two flows where fidelity is highest (SVDB $\rho = 0.83$, SMD $\rho = 0.74$), quality is lowest (0.058 and 0.036, both below the self-supervised arms). On the flow where quality is best (SWaT 0.588), fidelity is negative ($\rho = -0.20$). Three families, three architectures, one conclusion: ranking transfer and quality gain are decoupled regardless of how much of the teacher's ranking the student absorbs. Could a weak teacher explain the boundary instead? The frozen teacher's confirmation macro (0.265) sits below the linear floor (0.323), so teacher weakness is a natural confound for the no-transfer verdict. Three observations separate the interface account from the teacher-strength account. First, within the five sources where the teacher matches or beats the best self-supervised arm per flow, score distillation still adds only +0.018 mean VUS-PR (group means, exploratory stratification). Second, the TimesFM extension provides a same-flow natural experiment. On SMD the two teachers differ fivefold in zero-shot strength (TimesFM 0.382 versus Chronos 0.069), yet their distilled students land within 0.025 of each other (0.030 versus 0.055) and both far below the stronger teacher. Third, the capacity sweep shows the deficit shrinking smoothly with student size and closing only near six million parameters, far beyond the edge budgets at stake. Teacher strength modulates the ceiling. Within the

tested boundary, the interface is the binding constraint. These results are included in the artifact package.

7.2 A Transferability Screening Protocol

The controls above were run as science, but they double as an engineering tool. Before committing a TSFM teacher to a distillation pipeline, a deployer can screen transferability in three cheap steps, each priced in runs. Step one is an information probe. Distill once with time-shuffled teacher scores on a handful of streams (two seeds, about forty runs in our protocol). If real and shuffled scores produce indistinguishable students, the score interface carries nothing beyond the score marginal, and the teacher earns no place in the loop. Step two is a capacity probe. Repeat the score-distilled and self-supervised pair at three student sizes (about forty-five runs). A deficit that survives the largest size is an interface property, not a capacity artifact. Step three is an interface alternative. Swap the target to the teacher's forecasts and compare against plain self-supervision (about thirty runs). If neither exceeds self-supervision, the teacher can be retired from training as well as from inference. The full screen costs about 115 runs, roughly one fifth of the confirmation budget, and every step reuses cached teacher outputs. We present it as a screening heuristic derived from this study's controls. It is not a universal guarantee.

7.3 Deployment Considerations

Table 8 condenses our results into hypothesis-generating observations for deployment planning; it is conditional by design and is *not* a validated decision procedure. Three practitioner questions frame the table: (i) Can the detector be distilled offline, and how large is the artifact that must reach the device? (ii) Does the teacher need to accompany the deployment (it does not, for any student arm)? (iii) Which anomaly morphology is expected on the target stream, and does the chosen arm's failure mode match it? Each table row carries its failure mode; margins between neural arms are small and not significant.

7.4 Threats to Validity

We group threats into internal (identification and statistics), external (generalization), and construct (measurement) validity; the preregistration revisions R1–R17 and their timing are part of the internal-validity record. (1) The distillation signal exists only on normal-only training segments; behavioral transfer under real (non-synthetic) anomaly training data remains untested; the synthetic-anomaly control (R8) provides initial evidence. (2) Teacher-family checks cover three families. TimesFM 2.5 (decoder-only) spans ten flows and replicates the no-advantage verdict with high ranking fidelity (0.70). Moirai-1.1-R-base (masked-reconstruction, five flows under R16) repeats the pattern with fidelity up to 0.83. Three TimesFM streams remain structurally unscorable within the locked context. (3) The preregistration designated causal Matrix Profile and LOF as auxiliary baselines. Revision R11 added three zero-tuning classical baselines (robust-z, MA-residual, LOF-window) under the same scoring protocol; causal Matrix Profile remains unexecuted, as its correct causal implementation

Table 8 Practitioner selection table (exploratory; conditional on anomaly morphology and budget; see Section 5.4).

Condition	Observed candidate	Failure mode / caveat
Level or slow-regime anomalies; zero model, zero storage, zero tuning	MA-residual (causal moving average, L=128)	Confirmed near the top of the held-out macro (0.325); blind to short micro-transients inside the averaging window
Level/regime-shift anomalies; storage $\geq$ a few KB; offline reference OK	PCA reconstruction (fixed configuration)	Catastrophic on temporal/persistence anomalies (e.g., 0.007 on GHL); reference update needed under drift
Tightest latency budget; any morphology except micro-transients	Linear predictor + residual	Median quality below several neural arms (LINEAR dev macro 0.248 vs. B1-TCN 0.253, B2-TCN 0.251); window anomalies near stream start invisible
Temporal and persistence anomalies; tightest size budget after INT8	B1 (self-supervised) PatchMLP-lite INT8	Exploratory dev-source niche (SWaT 0.568 vs. 0.446 PCA; single-source observation); occupancy edge over C-PM is matched by zero-tuning robust-z
Small, smooth-deviation anomalies on short streams (exploratory; not validated on held-out sources)	Score/ranking-distilled student (C-PM)	Among the weakest arms on held-out sources (macro 0.250; the score arm A-PM is lower at 0.212); margins over B1 are small and not significant; loses the occupancy niche to B1-TCN (B1-TCN 1.000 vs. C-PM 0.560 on occupancy)

cost remains out of proportion to the question once three classical controls are present. (4) The confirmation set has nine sources. With flow-level bootstrap at 95%, the observed CI half-widths are 0.05 to 0.10 for neural-arm contrasts, so the minimum detectable effect is roughly 0.10 to 0.19 VUS-PR on the macro scale. The study rules out differences of that size and larger, not smaller ones, and all arm-level comparisons are non-significant at this sample size. Every no-advantage claim in this paper is bounded by that margin. (5) VUS-PR with default buffering is the primary metric; thresholded operating points and alerting dynamics are out of scope. (6) The preregistration was amended five times before any confirmation-set run (R1–R5) and three more times for exploratory controls (R6–R8), each disclosed with its trigger and hash (Appendix A); late amendments based on dev-only evidence are identified as such.

(7) On-device measurements on a Raspberry Pi 4B cover latency, memory, and a ten-minute sustained-throughput check with no thermal throttling (1800 MHz throughout, peak 55∘C). Energy profiling remains future work, and desktop single-thread figures remain the floor for every other configuration. (8) An audit of the training-segment splitter found one development source (GECCO) whose fallback split places 1.77% anomalies in its training segment; excluding it raises every arm's development macro by 0.006 to 0.013 with the ordering unchanged, and the normal-only descriptions are otherwise accurate (revision R16).

8 Conclusion

We asked what a frozen TSFM teacher, what distillation actually transfers, and whether the context asymmetry drives the outcome. The answers delineate a boundary rather than a verdict. First, within the tested boundary, the interface decides what transfers. The teacher's *score* interface transfers almost none of its ranking behavior (median $\rho = 0.184$, sharply bimodal under anomaly exposure) and yields no statistically detectable quality benefit. All preregistered confirmatory tests are non-significant, and the study rules out large differences only, with CIs extending to $+0.10$ and $+0.19$ for the two favorable comparisons. The teacher's *forecast* interface recovers most of the gap on development flows and is equivalent to self-supervision on held-out sources, a development-stage preference that did not reach significance there. Second, the failure is attributable, not anecdotal. Shuffled-teacher, equal-context, anomaly-exposure, and second-teacher-family controls jointly point to the score interface as the most consistent explanation within the tested teachers, capacities, and training conditions, without excluding those factors in general. Third, the deployment case is settled independently of the quality question. Students are 200–1500$\times$ cheaper per point than the teacher on trained artifacts, and quantized PatchMLP students meet sub-millisecond, sub-megabyte budgets at every tested channel count. The capacity sweep adds the flip side. Score distillation closes its deficit to self-supervision only near six million parameters, whose exported artifact measures 23.2 MB (5.8 MB quantized), so within edge budgets the interface remains the binding constraint. The contribution is a disciplined, reproducible boundary delineation with a mechanism analysis, an interface comparison that redirects future work toward forecast targets, and a conditional selection table for practitioners.

CRediT authorship contribution statement

Yao Zhang: Conceptualization, Methodology, Software, Investigation, Data curation, Writing – original draft, Visualization. **Yanling Li:** Supervision, Resources, Writing – review & editing. **Yichen Wang:** Data curation, Writing – review & editing.

Declaration of generative AI in the writing process

During the preparation of this work the corresponding author used an AI assistant (ZCode, GLM) for code development, experiment execution, data analysis, and drafting support. All experimental decisions followed a preregistered protocol (v1.0 with

append-only revisions R1–R17, released with the artifacts). The authors verified the results and take full responsibility for the content of the published article.

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Competing interests

The authors declare no competing interests.

Data and code availability

All code (teacher scoring, student training, export/timing, statistics), the preregistration with its revision log (including the full SHA-256 hash chain of Appendix A), per-run results, and the teacher-score cache metadata will be released in a public repository (GitHub with an archived Zenodo DOI) upon acceptance, and the complete artifact package accompanies the submission. Benchmark data are public (TSB-AD-M, TSB-UAD, NAB, and the original releases of SMD, PSM, MSL/SMAP, SWaT, HAI 2.2, MetroPT3, WADI, J-D, EMG, GHL, GECCO, Exathlon, Opportunity, Genesis, Daphnet, MITDB, SVDB). NAB window labels follow the official combined label file.

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The teacher-score cache and the detection-interface configuration were developed within a companion research effort and are reused with disclosure (Section 3.4).

A Preregistration Revision Log (Hash Chain)

Each revision is append-only; the confirmation data were untouched until after R5. Table 9 lists all eighteen revisions with their registered hashes. Revisions R1–R5 were registered before any confirmation-set run. R6–R8 are exploratory controls registered post-hoc (R6) or pre-run for post-hoc experiments (R7, R8), as flagged in the trigger column. R9 registered a third-teacher (Moirai) control that stayed unexecuted until R16 carried it out on five flows. R10 registers the submission-hardening controls, covering a capacity-interface sweep registered before its runs, a second-teacher full-flow extension registered mid-execution before any result was inspected, and a post-hoc stratified re-analysis. The preregistration text, this hash chain, and all per-run results are part of the artifact package. The chain evidences disclosure transparency. Full-length hashes and file timestamps are in the artifact package, and no external timestamping authority is claimed.

B Reproduction Details

Table 10 summarizes the reproduction configuration.

C Deployment-Cost Matrix on Trained Artifacts

Table 11 reports the full per-configuration cost matrix: three architectures at six channel counts (1, 9, 38, 66, 116, 248, each realized on a real flow of that dimensionality), FP32 and dynamic-INT8 variants, five timing rounds each. The timing artifacts are case-C students retrained under the confirmation recipe for this matrix; latency depends on the computation graph (architecture, dimension, precision), not on weight values, and the quality-drift column verifies that the deployed artifacts retain detection behavior. The end-to-end protocol (stride-1 windows over full test splits, single-thread, batch 1) was additionally measured on two development flows (NAB cpu, SMD) and one held-out confirmation flow (Daphnet): PatchMLP 0.069–0.084 ms p50, TCN 0.329–0.511 ms p50.

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Table 9 Preregistration revision log (hash chain). Each row records one append-only revision of the preregistration document; the hash column gives the first 16 hex characters of the SHA-256 digest of the registered file.

Rev.	Date/Time	SHA-256 (first 16 hex)	Trigger
v1.0	2026-10-03 00:23	315881ad7321641f	initial freeze, before any test metric
R1	2026-10-03 01:40	7e6f55158c350fb9	smoke-test metadata corrections
R2	2026-10-03 12:13	14f2dea01a2fcf07	WADI format, NAB label, ordering fixes
R3	2026-10-04 09:17	ed9ab91e219d26f2	AB-arm deviation; two-arm confirmation amendment
R4	2026-10-04 21:02	1948c19c4ff8ff2d	C9 replacement, CPU execution mode
R5	2026-10-04 22:50	6afbb2cc0720e685	five-arm confirmation amendment; MP/LOF non-execution disclosure
R6	2026-10-06 09:36	69177d7d3ae715b8	shuffled-teacher exploratory control (recorded post-hoc)
R7	2026-10-06 21:17	d9bacd89964b7c65	interface ablation at teacher context length (recorded pre-run)
R8	2026-10-07 22:37	ab88763bbd684611	teacher-on-anomaly distillation control (recorded pre-run)
R9	2026-10-08 23:33	d7f5d4e347cbc308	Moirai third-teacher control (registered pre-run; not executed)
R10	2026-10-13 01:08	9a8a606d3a3686e5	submission-hardening controls: second-teacher full-flow, capacity sweep, stratified macro
R11	2026-10-17 09:21	74143d38e48c8184	classical baselines (robust-z, MA-residual, LOF-window) and presentation hardening
R12	2026-10-17 09:37	f7d87930c89216b5	erratum: R11 timing-evidence wording corrected
R13	2026-10-17 10:04	8284d2bac26f3625	capacity sweep extension to 5.8M parameters (pre-run)
R14	2026-10-17 22:58	0f3fd5a2b282210a	classical-baseline latency measurement and cost-plane update (pre-run)
R15	2026-10-18 01:42	30ba2d792c9cff3f	on-device sustained-throughput and teacher-latency measurements (pre-run)
R16	2026-10-18 23:40	af2f5824507d6682	leakage audit disclosure, Moirai third-family execution, prospective tenth-source extension, stratified reanalysis
R17	2026-10-19 00:22	f61d9c351fb91c1f	R16 candidate-source erratum: CreditCard, CATSv2, TAO, LTDB

Table 10 Reproduction details.

Item	Value	Source
Student backbones	TCN-lite: 4-stage dilated causal conv (96/112/128/128, kernel 5), pool-4 head; PatchMLP-lite: patch 8 (PatchTST [16]), dim 128, 2 mixer blocks (MLP-Mixer [22])	artifact code
Parameters (66-ch instance)	TCN-lite 272,401; PatchMLP-lite 265,617; linear floor 8,383	counted at build
Training	Adam, lr 10^{-3}, batch 256, 30 epochs, window L=128, stride 2, per-flow	run manifests
Teacher scoring	context 512 (Chronos-Bolt) / 1024 (TimesFM 2.5); point forecast = median quantile (Chronos) or mean output (TimesFM)	cache metadata
Export	ONNX opset 17; dynamic batch; optional dynamic INT8	timing script
Run ledger	1,856 total: 1,270 preregistered (400 main grid + 420 auxiliary + 450 five-arm confirmation) + 586 post-hoc/controls and additions (90 A-PM confirmation addition + 40 shuffled-teacher R6 + 25 TimesFM subset + 30 interface ablation R7 + 30 anomaly-exposure R8 + 25 TimesFM full-flow R10 + 135 capacity-sweep R10 + 87 classical-baseline scoring R11 + 45 capacity-1024 R13 + 60 extension-arm R16 + 4 extension-evaluation R16 + 15 Moirai R16). Fig. 3 plots all runs with finite fidelity: 389 main-grid plus 420 auxiliary runs	per-run manifests

Table 11 Deployment-cost matrix on trained artifacts (case-C students, seed 0, M6 training protocol; idle desktop CPU, single-thread ONNX Runtime, batch 1, window L=128; median over 5 rounds of 50 warm-up + 500 timed windows, with round-to-round min–max of p95 in parentheses. The final column gives the INT8-minus-FP32 VUS-PR drift of the exported artifacts on each flow's test split.

Ch	Model	Prec	p50 (ms)	p95 (ms)	p95 range	MB	ΔVUS
1	LINEAR	FP32	0.009	0.009	0.009–0.013	0.00	ref
1	LINEAR	INT8	0.010	0.010	0.010–0.011	0.00	+0.000
1	PatchMLP	FP32	0.073	0.092	0.072–0.169	0.80	ref
1	PatchMLP	INT8	0.055	0.058	0.056–0.064	0.22	+0.000
1	TCN	FP32	0.346	0.452	0.437–0.473	0.97	ref
1	TCN	INT8	4.315	4.653	4.550–4.778	0.26	-0.000
9	LINEAR	FP32	0.009	0.010	0.009–0.011	0.01	ref
9	LINEAR	INT8	0.011	0.011	0.011–0.011	0.00	-0.002
9	PatchMLP	FP32	0.077	0.092	0.079–0.168	0.83	ref
9	PatchMLP	INT8	0.060	0.072	0.061–0.093	0.22	+0.005
9	TCN	FP32	0.335	0.414	0.386–0.432	0.98	ref
9	TCN	INT8	4.109	4.391	4.341–4.512	0.26	-0.009
38	LINEAR	FP32	0.012	0.013	0.013–0.019	0.02	ref
38	LINEAR	INT8	0.013	0.014	0.014–0.014	0.01	+0.000
38	PatchMLP	FP32	0.091	0.108	0.102–0.121	0.95	ref
38	PatchMLP	INT8	0.066	0.072	0.069–0.082	0.25	-0.000
38	TCN	FP32	0.363	0.429	0.416–0.444	1.04	ref
38	TCN	INT8	4.544	4.842	4.821–4.916	0.28	-0.001
66	LINEAR	FP32	0.016	0.016	0.016–0.020	0.03	ref
66	LINEAR	INT8	0.016	0.016	0.016–0.016	0.01	-0.066
66	PatchMLP	FP32	0.099	0.116	0.106–0.120	1.07	ref
66	PatchMLP	INT8	0.069	0.082	0.077–0.094	0.28	+0.074
66	TCN	FP32	0.388	0.474	0.450–0.508	1.09	ref
66	TCN	INT8	4.943	5.460	5.379–5.567	0.29	+0.048
116	LINEAR	FP32	0.023	0.024	0.024–0.025	0.06	ref
116	LINEAR	INT8	0.020	0.023	0.022–0.024	0.02	+0.000
116	PatchMLP	FP32	0.113	0.142	0.130–0.147	1.27	ref
116	PatchMLP	INT8	0.078	0.085	0.080–0.093	0.33	+0.000
116	TCN	FP32	0.438	0.531	0.521–0.537	1.19	ref
116	TCN	INT8	5.710	6.052	6.020–6.159	0.31	-0.002
248	LINEAR	FP32	0.047	0.049	0.046–0.057	0.13	ref
248	LINEAR	INT8	0.033	0.035	0.033–0.037	0.03	+0.009
248	PatchMLP	FP32	0.158	0.194	0.176–0.199	1.81	ref
248	PatchMLP	INT8	0.105	0.116	0.113–0.136	0.47	+0.097
248	TCN	FP32	0.561	0.653	0.634–0.769	1.44	ref
248	TCN	INT8	7.640	7.995	7.989–8.034	0.38	+0.002